

OMAR EL-SHAROUD

Software Engineer

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PROFESSIONAL PROFILE

Software Engineer with experience delivering enterprise business solutions on Microsoft Power Platform, Dynamics 365 and Azure, alongside full-stack web applications built in PHP, Node.js and React. At OGI, contributed to annual recurring billing automation, Dynamics customer-service enhancements, a warehouse ticketing system, a contract-import integration and the AssetTrack asset management platform, taking solutions from design through UAT to production with Azure DevOps CI/CD. Independently built and maintain the Dar UI Isra and Bayan Academy websites, including online registration and card payments. BSc (Hons) Software Engineering graduate with a strong grounding in the full SDLC, secure systems and testing.

TECHNICAL TOOLKIT

- **Microsoft Business Apps:** Microsoft Power Platform (Power Apps, Power Automate, Dataverse), Dynamics 365 Customer Service & Plugin Development, Model-Driven Apps, SharePoint Online
- **Cloud & DevOps:** Microsoft Azure (App Service, PostgreSQL, Key Vault), Microsoft Entra ID, Azure DevOps, CI/CD, Docker, Git/GitHub
- **Languages & Web:** Python, Java, PHP, JavaScript, SQL (MySQL, PostgreSQL), HTML, CSS, WordPress
- **Windows Ecosystem:** Windows OS Architecture, System Services, Frameworks, Security Features
- **DevOps & Tools:** Git, GitHub (Branching, Pull Requests, Version Control), Docker, Docker Compose, CI/CD Pipelines, Cisco Packet Tracer
- **Methodologies & Testing:** Agile (Scrum), Full SDLC Management, Unit Testing, Integration & System Testing, Advanced Debugging
- **Security & Compliance:** Data Hashing (Bcrypt, SHA), Secure Session Management, PDO Prepared Statements, GDPR Awareness
- **Delivery:** REST API Integrations, Billing & Finance Integrations, Solution Architecture, Requirements Gathering, UAT & Production Deployments, Technical Documentation & Handover

PROFESSIONAL EXPERIENCE

Software Engineer, OGI | 2026

Annual Recurring Billing | *Power Platform / Dynamics Developer*

Helped bring annual recurring billing to OGI's Dynamics environment, covering renewals, annual contract handling and pricing. Reworked the pricing logic and product synchronisation, updated the Power Automate flows and Dynamics plugins, and connected everything to the BillingBooth API, with JavaScript web resources on the front end. Also handled testing and deployment planning.

Technologies: Dynamics 365, Dataverse, Power Automate, BillingBooth API, JavaScript, Azure DevOps, Power Platform Solutions

IP Allocation Visibility in Dynamics | *Dynamics 365 Developer*

Built a way for customer service teams to see a customer's Fixed IPv4 and Routed Subnet allocations directly in Dynamics 365. Added the new fields, updated the forms and customer records, wrote the high-level design document, and tested the solution through to UAT and production.

Technologies: Dynamics 365, Dataverse, Model-Driven Apps, Solution Management, UAT Testing

Warehouse Ticketing System | *Power Platform Developer*

Designed and delivered an end-to-end ticketing system for the warehouse team: ticket submission, automatic numbering, confirmation emails, status updates and SharePoint integration. Wrote the test plans, deployed to UAT, then took it through the release pipeline to production, delivering the warehouse team's follow-up change requests along the way.

Technologies: SharePoint Online, Power Apps, Power Automate, Dataverse, Azure DevOps, CI/CD Pipelines

PSI Fusion Contract Import | *Power Platform Developer*

Built an integration that pulls thousands of PDF contract records from PSI Fusion into Dataverse using Power Automate, with a one-off bulk migration followed by automated daily imports of new and changed records. Includes duplicate prevention, metadata management and error handling, plus a model-driven app for finding and opening the documents.

Technologies: Power Automate, Dataverse, Model-Driven Apps, REST APIs, JSON

AssetTrack | *Full Stack Developer*

Developed AssetTrack, an internal platform for tracking engineering assets and certifications. Includes engineer, manager and admin roles, expiry tracking and certification management, with authentication and security controls built in.

Technologies: React, Node.js, Express, PostgreSQL, Microsoft Entra ID, Azure DevOps

AssetTrack Azure Migration | *Cloud & Application Developer*

Planned and drove the move of AssetTrack from local Docker infrastructure to Azure. Designed the Azure architecture, role-based access model, database migration approach and Entra ID authentication, and wrote the security documentation, VPN access requirements and deployment policies.

Technologies: Azure App Service, Azure PostgreSQL, Azure Key Vault, Entra ID, Docker, VPN & Networking

Dynamics 365 Plugin & Solution Development | *Dynamics Developer*

Developed and tested Dynamics plugins and solutions covering customer service records, pricing, product synchronisation, order processing and billing integrations, and released them through Azure DevOps pipelines as managed solutions into UAT and production.

Technologies: Dynamics 365, C# Plugins, Dataverse, Azure DevOps, Managed Solutions

Power Platform CI/CD & Environments | *Power Platform Developer*

Managed the solution lifecycle across Development, UAT and Production: deployments, connection references, environment variables, managed and unmanaged solutions, and Azure DevOps release pipelines.

Technologies: Power Platform, Azure DevOps, Connection References, Environment Variables

HMRC - Administrative Assistant | *Sept 2022 – Sept 2023*

- Managed high-volume customer queries, applying highly structured, analytical problem-solving methodologies mirroring software debugging workflows.
- Processed sensitive data in strict accordance with compliance frameworks and data protection legislation, mirroring backend security practices.
- Developed exceptional skills in process optimisation, systematic troubleshooting, and delivering accurate results under tight operational pressure.

TECHNICAL PROJECTS

Bayan Academy Website

- Built the official website and online registration system for Bayan Academy (Salam Centre, Cardiff) in PHP and MySQL, with a bilingual English and Arabic public site covering pathways, courses, timetable, fees calculator and FAQ.
- Integrated online card payments for registration and a family payment link for outstanding balances, with server-side pricing, webhooks that record payments even if the payer never returns, and automatic confirmation of each payment with the provider's API.
- Implemented the fee rules (per-term, annual and monthly Direct Debit options, family and multi-course discounts, promo codes) with a daily cron job for fee entries, payment reminders and an office digest.
- Created a staff area where the Director and teachers manage registrations, payments, promo codes, the course catalogue, emails and the site's wording without touching code.

Dar UI Isra Platform

- Developed and continue to maintain the official community platform using WordPress, acting as the sole technical lead for a digital hub serving a large, highly diverse user base.
- Integrated dynamic backend functionality, including automated prayer timetables, event management calendars, and secure donation gateways.

- Optimised the UI/UX for exceptional accessibility, mobile responsiveness, and technical SEO, ensuring critical information is easily accessible.

Dar Ul Isra Youth Halaqas Registration

- Built a multi-step online registration for the mosque's weekly Youth Halaqas (ages 7-18), collecting child details, parent/guardian and emergency contacts, medical information and photography consent, with children placed into separate age and gender groups.
- Supported registering several children in a single submission, a waiting list when groups are full, and a family-grouping preference.
- Integrated Stripe so card details are never handled by the mosque, with a payment link sent only once a place is confirmed.

Iqraa Quran School Registration

- Built the online registration form for the mosque's Quran classes, capturing up to three students per submission with their Quran reading level (Qaida through to Fluent), guardian and emergency contacts, medical details and emergency treatment consent.
- Added attendance and parental-responsibility acknowledgements with a typed digital signature, giving the school a complete, consistent record for each family.

WhatsApp AI Automation SaaS

- Built a multi-tenant SaaS platform using Node.js, Express, PostgreSQL, and Docker that connects seamlessly to WhatsApp Web to automate intelligent, context-aware replies powered by the Google Gemini API.
- Engineered persistent, Puppeteer-based session management with compressed database-backed archival, enabling flawless WhatsApp connectivity across container rebuilds without requiring user re-authentication.
- Designed a multi-group monitoring system featuring per-chat AI persona overrides, a one-tap web-scraping knowledge pipeline, and an automated availability calendar engine with iCal export functionality for autonomous scheduling.
- Containerised and deployed the entire infrastructure via Docker Compose to Hugging Face Spaces with automated CI/CD pipelines, featuring a mobile-responsive telemetry dashboard and secure JWT/Bcrypt authentication.

Align

- Architected and deployed a full-stack marriage compatibility platform featuring a psychometric assessment engine, real-time matching algorithm, and privacy-first photo reveal system built with React, Express 5, SQLite, and Prisma ORM.
- Designed and implemented a multi-journey psychometric assessment flow comprising 40+ adaptive question screens (single-select, multi-select, ranked-choice, points-distribution, timed-response) with conditional branching logic based on user demographics and real-time selections.
- Engineered a weighted compatibility matching algorithm with dealbreaker filtering, bidirectional scoring, and cold-start handling, backed by a normalised relational schema managing users, assessment responses, match pairs, and messaging logs.
- Built a staged photo privacy pipeline using server-side image processing to generate pristine, 30%-blurred, and silhouette variants, progressively revealing profile photos based on mutual match acceptance and journey completion milestones.
- Developed a real-time communication layer using WebSockets for instant messaging, with Agora RTC integration for in-app voice and video calling between matched users.
- Implemented a secure authentication system with JWT tokens, AES-256 encryption for sensitive PII fields, and a face-detection liveness verification flow using the face-api.js library.
- Containerised the application with a multi-stage Dockerfile and deployed to Fly.io with persistent SQLite volumes, automated CI/CD via GitHub Actions, and zero-downtime redeployment on push to main.

AccomFix

- Developed a full-stack web application utilising the LAMP stack (PHP, MySQL, Apache, HTML/CSS/JS) to streamline automated maintenance reporting and triage workflows within the student housing sector.

- Engineered a custom 'Link Code' multi-tenancy architecture, guaranteeing secure, frictionless onboarding and total data isolation between independent landlords and tenant clusters.
- Implemented a priority-based ticketing workflow featuring secure multipart file uploads for visual evidence alongside a real-time, AJAX-driven live messaging system.
- Designed a role-based administrative dashboard leveraging SQL aggregate functions for real-time issue analytics, status management, and automated CSV data exportation.
- Enforced strict web security best practices, deploying Bcrypt for password hashing and PDO prepared statements to completely mitigate SQL injection vulnerabilities.

CampusTasker

- Engineered a localised peer-to-peer web marketplace using PHP, MySQL, and Vanilla JavaScript designed to connect community residents with university students for local tasks.
- Integrated third-party spatial APIs (Postcodes.io and Leaflet.js) using PHP cURL to build an interactive mapping system that dynamically geocodes locations and calculates real-time distances.
- Designed a secure, relational database schema implementing robust error-handling for API timeouts and comprehensive data validation.
- Built an asynchronous messaging system and dynamic marketplace filters using the JavaScript Fetch API, enabling real-time DOM polling and UI updates without full-page reloads.
- Applied the POST/Redirect/GET (PRG) architectural pattern to ensure secure state management and prevent duplicate database entries during form submissions.

QuizCraft

- Developed a dynamic, custom MVC-architected web application enabling users to build, publish, and evaluate highly interactive educational quizzes.
- Designed a normalised relational database schema using PDO to securely manage users, complex relational quiz structures, dynamic multiple-choice arrays, and submission logs.
- Engineered advanced quiz flow logic, including automated grading engines and conditional branching paths based on real-time user selections.
- Built a highly interactive front-end utilising Vanilla JavaScript for real-time DOM manipulation, allowing users to dynamically alter form elements seamlessly.

Mental Health Support Platform

- Oversaw the entire development lifecycle from initial requirement gathering through to deployment, managing task delegation within an Agile framework.
- Developed a secure web application with robust user authentication frameworks, session tracking, and encrypted database hashing.
- Conducted rigorous unit testing on booking algorithms and security modules to guarantee data integrity and system stability.

EDUCATION

BSc (Hons) Software Engineering, Cardiff Metropolitan University | Sept 2023 – Graduate (06/2026)

- Relevant Modules: Operating Systems & Architecture, Advanced Object-Orientated Programming, Database Management Systems, Systems Analysis & Design.

A-Levels & Equivalents, Cardiff High School | Sept 2018 – June 2022

- Subjects Covered: Mathematics, ICT, Physics, Chemistry, Biology, Design & Technology, Business Studies.

LEADERSHIP & VOLUNTEERING

Youth Programme Coordinator & Lead Mentor | Community Hub (Sept 2024 – Present)

- Designed and implemented a comprehensive three-year educational and development curriculum tailored for distinct youth demographics.
- Facilitated weekly interactive sessions focusing on strategic analysis, character building, historical studies, and community ethics.

- Mentored young adults through complex social topics, actively fostering personal accountability, resilience, and critical thinking skills.
- Adapted advanced historical texts into engaging, accessible narratives while successfully coordinating project logistics and managing volunteer teams.

Scout Leader | *The Scouts Association (Ongoing)*

- Organised and led large-scale camps and expeditions, taking full responsibility for risk assessments, safety protocols, and complex logistics.
- Mentored youth members to foster teamwork, resilience, and practical problem-solving capabilities.
- Successfully transferred these leadership skills into technical settings by guiding software development teams, setting clear objectives, and delegating development tasks effectively.

LANGUAGES & ADDITIONAL INFORMATION

- **Languages:** English (Bilingual), Arabic (Bilingual)
- Full UK Driving License (since Oct 2021).